

Epic 5: Application Building

Story 1: Frontend Development

The frontend of the AI Flood Prediction System is developed using HTML, CSS, and JavaScript within the Flask framework. Flask organizes the user interface by storing HTML pages in the `templates/` folder and static resources such as CSS, JavaScript, and images in the `static/` folder. This structure ensures a clean, responsive, and user-friendly web application.

Templates Folder (`templates/`)

The `templates/` folder contains all the HTML pages used by the application.

- `home.html` – Displays the home page with an introduction to the AI Flood Prediction System and its objectives.
- `index.html` – Provides a user-friendly form for entering weather parameters such as rainfall, cloud cover, humidity, and temperature. The entered data is submitted to the Flask backend for prediction.
- `chance.html` – Displays an alert message when the model predicts a high probability of flooding, enabling users to take necessary precautions.
- `no_chance.html` – Displays a safe status message when the model predicts no flood risk for the given weather conditions.

Static Folder (`static/`)

The `static/` folder contains the frontend assets required to enhance the application's appearance and functionality.

- `main.css` – Defines the visual design, layout, colors, typography, and responsive styling of the web pages.
- `main.js` – Provides client-side functionality such as form validation, button interactions, and user interface enhancements.

Benefits

- Organizes the application using Flask's standard project structure.
- Creates a clean and responsive user interface.
- Improves user experience through interactive web pages.
- Separates presentation logic from backend functionality, making the application easier to maintain and extend.

Outcome

- Designed responsive HTML pages for user interaction.
- Implemented CSS styling for an attractive interface.
- Added JavaScript for validation and interactivity.
- Successfully prepared the frontend for integration with the Flask backend and machine learning model.

Figure 18: Frontend structure of the AI Flood Prediction System showing the templates/ and static/ folders.

